



COGNITION & MOOD

Semax

Semax – 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Methionyl-glutamyl-histidyl-phenylalanyl-prolyl-glycyl-proline
SYNONYMS / ALIASES	Semax; ACTH(4-10) Pro-Gly-Pro analog
CAS NUMBER	80714-61-0
MOLECULAR FORMULA	C37H51N9O10S
MOLECULAR WEIGHT	813.92 g/mol
AMINO ACID SEQUENCE	MEHFPGP (7 residues; H-Met-Glu-His-Phe-Pro-Gly-Pro-OH)
SOURCE	Synthetic (solid-phase peptide synthesis); designed in Russia as a stabilized analog of the ACTH(4-7) fragment with a C-terminal Pro-Gly-Pro tripeptide extension for enzymatic stability

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 28 days in bacteriostatic water
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Methionine residue is oxidation-sensitive; protect from prolonged air exposure. The C-terminal Pro-Gly-Pro extension confers significant resistance to proteolytic degradation.

STRUCTURAL NOTES

Semax is a synthetic heptapeptide (7 amino acids) consisting of the ACTH(4-7) tetrapeptide fragment (Met-Glu-His-Phe), corresponding to residues 4-7 of native adrenocorticotrophic hormone (ACTH), extended at the C-terminus by the tripeptide Pro-Gly-Pro. The Pro-Gly-Pro extension was added to confer resistance to proteolytic cleavage relative to the native ACTH(4-10) fragment. Semax was developed by the Institute of Molecular Genetics of the Russian Academy of Sciences.

SELECTED PEER-REVIEWED REFERENCES

1. Ashmarin I.P. et al. - *Nootropic analog of adrenocorticotrophic hormone 4-10: Semax (15 years of experience in development and study)* - Zhurnal Vysshei Nervnoi Deyatelnosti, 1997
2. Kaplan A.Y. et al. - *Synthetic ACTH analogue Semax displays nootropic-like activity in humans* - Neuroscience Research Communications, 1996
3. Medvedeva E.V. et al. - *The peptide semax affects the expression of genes related to the immune and vascular systems in rat brain focal ischemia: genome-wide transcriptional analysis* - BMC Genomics, 2014 - DOI: 10.1186/1471-2164-15-228

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.